



Engineer to Excel

SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME
DSA0115 Object Oriented Programming with C++
for System Level Programming

OPERATOR OVERLOADING

SIMATS
ENGINEERING

SIMATS Online Programming Lab

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QUESTIONS 5 / 19 completed

Plus

PLUS

Write a C++ program to overload the + operator to add two complex numbers.

PROGRAM EDITOR

C++

Run

Save

```
1 // Number class
2
3 #include <iostream>
4 int value;
5 public:
6     Number(int v = 0) {
7         value = v;
8     }
9     Number operator+(Number obj) {
10         Number temp;
11         temp.value = value + obj.value;
12         return temp;
13     }
14     void display() {
15         cout << "Result = " << value << endl;
16     }
17 };
18 int main() {
19     Number n1(10), n2(20);
20     Number n3;
21     n3 = n1 + n2;
22     n3.display();
23     return 0;
24 }
```

OUTPUT

Result = 30

INPUT

10
20

QUESTIONS 5 / 19 completed

Increment

INCREMENT

Write a C++ program to overload the ++ operator to increment a variable.

PROGRAM EDITOR

C++

Run

Save

```
2 using namespace std;
3 class Number {
4     int num;
5 public:
6     void getData() {
7         cout << "Enter a number: ";
8         cin >> num;
9     }
10    void operator++() {
11        ++num;
12    }
13    void display() {
14        cout << "Value after increment = " << num << endl;
15    }
16 };
17 int main() {
18     Number n;
19     n.getData();
20     ++n;
21     n.display();
22     return 0;
```

OUTPUT

```
Enter a number: Value after increment = 16
```

INPUT

15

QUESTIONS 5 / 19 completed

<<

<<

Write a C++ program to overload the << operator to print the contents of a user-defined class.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Student {
4     int id;
5     string name;
6 public:
7     Student(int i, string n) {
8         id = i;
9         name = n;
10    }
11    friend ostream& operator<<(ostream &out, Student s) {
12        out << "ID : " << s.id << endl;
13        out << "Name : " << s.name << endl;
14        return out;
15    }
16 };
17 int main() {
18     Student s(101, "Sravanthi");
19     cout << s;
20     return 0;
21 }
```

OUTPUT

```
ID : 101
Name : Sravanthi
```

INPUT

```
101
Sravanthi
```

QUESTIONS 5 / 19 completed

Equal

EQUAL

Write a C++ program to overload the == operator to compare two objects of a user-defined class.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Number {
4     int value;
5 public:
6     Number(int v = 0) {
7         value = v;
8     }
9     bool operator==(Number obj) {
10         return value == obj.value;
11     }
12 };
13 int main() {
14     Number n1(25), n2(25);
15     if (n1 == n2)
16         cout << "Both are Equal";
17     else
18         cout << "Not Equal";
19     return 0;
20 }
```

OUTPUT

Both are Equal

INPUT

10
10

QUESTIONS 5 / 19 completed

{} ▾

{} ▾

Write a C++ program to overload the [] operator to access the elements of an array using an index.

PROGRAM EDITOR

C++ ▾

Run

Save

```
3 class Array {
4     int arr[5];
5 public:
6     void getData() {
7         cout << "Enter 5 array elements:\n";
8         for (int i = 0; i < 5; i++) {
9             cin >> arr[i]; }
10    int operator[](int index) {
11        if (index >= 0 && index < 5)
12            return arr[index];
13        else {
14            cout << "Invalid Index!" << endl;
15            return -1; }
16    int main() {
17        Array a;
18        int index;
19        a.getData();
20        cout << "Enter index to access (0-4): ";
21        cin >> index;
22        cout << "Element at index " << index << " = " << a[index] << endl;
23        return 0;
```

OUTPUT

```
Enter 5 array elements:
Enter index to access (0-4): Element at index 2 = 30
```

INPUT

```
10
20
30
50
58
2
```